Task 1:

Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

Steps Performed

1. I opened Notepad as Administrator on my Windows computer.
2. I opened the hosts file located at C:\Windows\System32\drivers\etc\hosts.
3. I added the following entry to map Spotify's domain to an incorrect IP address.

192.0.2.10 www.spotify.com

4. I saved the hosts file.
5. I opened Command Prompt.
6. I executed the following command:

ping www.spotify.com

7. I checked the output to see whether the domain resolved to the incorrect IP address.
8. The ping request may fail because the IP address is reserved for documentation and is not a real public Spotify server.

Example output:

Pinging www.spotify.com [192.0.2.10] with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

The exact error may differ. You might see Destination host unreachable or another message depending on your network.

Result: The hosts file was modified to map www.spotify.com to an incorrect IP address. The ping test demonstrated how an incorrect local hostname mapping can prevent successful communication.

Important: After the practical, remove the added line from the hosts file and save it to restore normal name resolution.

Task 2:

Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

Steps Performed

1. I opened Cisco Packet Tracer and added one DHCP server, one switch, and one client PC.
2. I connected the server and client PC to the switch using network cables.
3. I assigned the DHCP server a static IP address of 192.168.1.2 with subnet mask 255.255.255.0.
4. I opened the DHCP service settings on the server and turned the service On.
5. I created a DHCP pool with the start IP address 192.168.1.100 and subnet mask 255.255.255.0.
6. I intentionally configured the default gateway as 192.168.1.254, which was incorrect for this example network.
7. I opened the client PC and selected Desktop → IP Configuration → DHCP.
8. I checked that the client received an IP address and the incorrect default gateway from the DHCP server.
9. I opened Command Prompt on the client PC and used ipconfig to verify the assigned IP address, subnet mask, and default gateway.
10. I used the ping command to test connectivity to the correct router gateway, if one was configured.
11. To identify the problem, I compared the gateway received from DHCP with the actual IP address of the router's LAN interface.
12. I fixed the issue by changing the DHCP pool's default gateway to the correct router IP address, for example 192.168.1.1.
13. I renewed the client's DHCP configuration by selecting DHCP again or using ipconfig /release followed by ipconfig /renew on a Windows client.
14. I checked the gateway again and tested connectivity to the router and an external network, if available.

Result: I simulated an incorrect default gateway in the DHCP configuration, verified the gateway assigned to the client, and corrected the DHCP pool settings to restore proper network connectivity.

Task 3:

Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

Steps Performed

1. I checked the network connection and confirmed that devices could access external IP addresses but could not open websites using domain names.
2. I opened Command Prompt on the client computer.
3. I used the nslookup command to test DNS resolution.

   nslookup www.google.com
4. I checked the output and found that the DNS server was not resolving the domain name correctly.
5. I used the following command to check which DNS server was configured on the computer.

   ipconfig /all
6. I verified the DNS server IP address and checked whether the DNS server was reachable.
7. I tested another DNS server using the following command.

   nslookup www.google.com 8.8.8.8
8. I identified that the problem could be caused by an incorrect DNS server address or a DNS service configuration issue.
9. I corrected the DNS server address in the network settings or DHCP configuration to point to a working DNS server.
10. I cleared the local DNS cache using the following command.

    ipconfig /flushdns
11. I ran nslookup www.google.com again to verify that the domain name resolved to an IP address.
12. I opened the website in a browser to confirm that name resolution was working correctly.

Result: The DNS troubleshooting process helped identify a possible DNS server configuration problem. After correcting the DNS settings, domain names could be resolved and websites could be accessed normally.

Task 4:

A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

Answer:

Steps Performed

1. I checked the client device's IP configuration and found that it had received an APIPA address beginning with 169.254.x.x.
2. I checked whether the DHCP server was powered on and connected properly to the network.
3. I verified that the network cables and switch ports were working correctly.
4. I checked whether the DHCP service was enabled on the server.
5. I verified that the DHCP pool had available IP addresses and that the configured IP range and subnet mask were correct.
6. I checked whether the client and DHCP server were in the same network. If they were on different networks, I checked whether a DHCP relay was configured on the router.
7. I checked the DHCP server's default gateway and other network settings for configuration errors.
8. I renewed the client's IP configuration. On a Windows client, I used these commands:
9. ipconfig /release

ipconfig /renew

10. I checked the client's IP address again using:

ipconfig /all

11. I verified that the client received a valid IP address from the configured DHCP pool instead of an APIPA address.

Possible Causes

- DHCP server is switched off or the DHCP service is disabled.
- Network cable is disconnected or the switch port is faulty.
- DHCP pool has no available IP addresses.

- DHCP pool settings are incorrect.
- Client and server are on different subnets without a DHCP relay.
- DHCP requests are blocked by a firewall or network configuration issue.

Result: I identified the possible causes of the APIPA address and documented the troubleshooting process to restore DHCP connectivity and obtain a valid IP address.

Task 5:

Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Common DNS and DHCP Issues:

1. DNS Issues

No.	Issue	Symptoms	Possible Causes	Recommended Solution
1	DNS resolution failure	Websites do not open by domain name but may work by IP address.	Incorrect DNS server address or DNS server unavailable.	Configure a working DNS server and test using nslookup.
2	Incorrect DNS record	A domain resolves to the wrong IP address.	Incorrect A record or hosts file entry.	Correct the DNS record or remove the incorrect hosts file entry.
3	DNS server not responding	nslookup shows a timeout or server failure.	DNS service is stopped, network connectivity is lost, or port 53 is blocked.	Restart the DNS service and check network connectivity and firewall rules.

2. DHCP Issues

No.	Issue	Symptoms	Possible Causes	Recommended Solution
1	APIPA address assigned	The device receives an IP address beginning with 169.254.x.x.	DHCP server is unavailable or the client cannot reach it.	Check the DHCP service, cables, switch ports, and network configuration.
2	Incorrect default gateway	The device can communicate locally but cannot access other networks.	Incorrect gateway configured in the DHCP pool.	Correct the default gateway in the DHCP settings and renew the client's IP address.
3	DHCP pool exhausted	New devices cannot obtain an IP address.	All available addresses in the DHCP pool are assigned.	Expand the address pool if appropriate or release unused DHCP leases.

Conclusion: DNS issues affect domain name resolution, while DHCP issues affect automatic IP address assignment and network configuration. Checking settings, testing connectivity, and correcting server configurations can help resolve these problems.